1. Project Overview

Urban Logistics & Crowdsourced Courier Network is a platform where everyday commuters (freelance couriers) can pick up and deliver packages along their existing routes. The system manages the matching of senders, the movement of goods through urban hubs, and the financial clearing for labor.

2. Nine Core Features (Requirement)

- **Multi-Persona User Architecture**: A single account system where a user can switch between **Sender** and **Freelance Courier** roles, each with distinct profiles and permissions (using EER specialization).
- **Dynamic Pricing Engine**: A stored procedure that calculates delivery fees based on weight, distance, and "Urgency Factor," adjusting prices based on current driver availability in a specific zone.
- **Real-Time Milestone Tracking**: A historical tracking table that logs every status change (e.g., *Pending, Hub-Sorting, Out-for-Delivery, Completed*) with high-precision timestamps.
- **Automated Sorting-Hub Allocation**: Logic that automatically assigns a parcel to the nearest physical "Urban Hub" based on the sender's geographic coordinates.
- **Crowdsourced Labor Incentives**: A system that applies "Peak Hour" or "Weather Multipliers" to courier earnings, rewarding drivers for taking deliveries during high-demand periods.
- **Trust-Score & Reputation System**: A weighted feedback mechanism where ratings from both senders and recipients influence a courier’s "Trust Index," affecting their eligibility for high-value deliveries.

- **Escrow-Style Payment Management**: A transaction-safe system where funds are marked as HELD upon order placement and only RELEASED to the courier's ledger when the parcel status hits DELIVERED.
- **Zone-Based Geo-Fencing**: Constraints that limit "Standard" couriers to specific urban zones while requiring "Certified" couriers for inter-city or high-risk items (e.g., fragile or perishable goods).
- **Admin Analytics Dashboard**: SQL Views that aggregate data to show high-traffic delivery corridors, average delivery times per zone, and total platform revenue.

EER design decisions mapped to your 9 features:

ISA Specialization (Feature 1 — Multi-Persona): USER is the superclass with an *overlapping, total* participation constraint — a single account can be both SENDER and FREELANCE_COURIER simultaneously, with distinct attribute sets per role.

Weak Entities: TRACKING_MILESTONE (depends on DELIVERY_ORDER for identity) and COURIER_LEDGER (depends on TRANSACTION) are modeled as weak entities with composition arrows, reflecting Feature 3 and Feature 7.

Dynamic Pricing & Incentives (Features 2 & 5): DELIVERY_ORDER carries UrgencyFactor and QuotedFee. The INCENTIVE_MULTIPLIER entity is linked to ZONE, storing MultiplierType (PeakHour, Weather) and a Factor decimal applied to courier earnings.

Escrow Payment (Feature 7): TRANSACTION holds Status (HELD → RELEASED), HeldAt, and ReleasedAt timestamps. It feeds COURIER_LEDGER only when the corresponding TRACKING_MILESTONE reaches DELIVERED.

Zone Geo-Fencing (Feature 8): ZONE.ZoneType and ZONE.IsInterCity plus FREELANCE_COURIER.CertificationLevel together enforce which couriers can operate where, and PARCEL.RequiresCertified gates high-risk items.

Trust Score (Feature 6): RATING has both Score and WeightedContrib fields—the weighted average rolls up into FREELANCE_COURIER. TrustIndex via a stored procedure or trigger.

Admin Analytics (Feature 9): SQL Views would join DELIVERY_ORDER, URBAN_HUB, ZONE, TRACKING_MILESTONE, and TRANSACTION — all the data is available through these normalized relationships.